

Direct Integrals

bounded operators on Hilbert space H

Let $M \subset B(H)$. Then its **commutant** is the centraliser

$$M' = \{T : Tm = mT \ \forall m \in M\} \subset B(H).$$

automatically closed, hence a C^* -algebra

A sub- $*$ -algebra $A \subset B(H)$ is a **von Neumann algebra** if $A'' = A$. A is called a **factor** if its centre $Z(A) = A \cap A' = \mathbb{C}I$.

Examples:

- $A = B(H)$ is a Type-I factor;
- $A = \mathbb{C}I$;
- $A = \text{Diag} \subset M_2(\mathbb{C}) = B(\mathbb{C}^2)$ is a von Neumann algebra but not a factor.

Let $A \subset \bar{A}^n \subset \bar{A}^s \subset \bar{A}^w$ be the closures of A under the operator norm, strong, and weak topology, respectively

Thm (von Neumann Bicommutant): Let $A \subset B(H)$ be a unital $*$ -subalgebra. Then

$$\bar{A}^s = \bar{A}^w = A'',$$

i.e. the closure is the **von Neumann algebra** generated by A .

We have $M \subset M''$, so $M''' \subset M'$ and hence $M'' = M'''$. If M is the smallest subalgebra containing M then $M' \in M'$, so $M'' \subset M' = M$. Since M' is unital, $M'' = M$.

Lemma: A von Neumann algebra is generated by its unitary elements.

Let $\mathcal{A}_R$ be the $\mathbb{R}$ -vector space of self-adjoint elements, so $\mathcal{A} = \mathcal{A}_R + i\mathcal{A}_R$.

Let $T \in \mathcal{A}_R$, $f \in S(\mathbb{R})$ s.t. $f = \text{id}_R$ on $\text{Spec } T$. Then

$$T = f(T) = \int_{\mathbb{R}} f(\gamma) e^{2\pi i \gamma T} d\gamma.$$

unitary

Representations

A unitary representation π of G is a **factor representation** if the von Neumann algebra $VN(\pi)$ generated by $\pi(G) \subset B(V_\pi)$ is a factor.

Note that irreducible representations are factor, by Schur's lemma.

We call π_1, π_2 **quasi-equivalent** if $\exists$ a C^* -isomorphism ϕ s.t.

$$G \begin{matrix} \nearrow VN(\pi_1) \\ \searrow \downarrow \phi \\ VN(\pi_2) \end{matrix}$$

Lemma: Two irr. unitary representations are quasi-equivalent iff they are unitarily equivalent.

Hilbert Integrals

Let $\mathcal{A}$ be a σ -algebra on X , $\tilde{H} \rightarrow X$ a Hilbert bundle $\zeta_i \in H^0(\tilde{H})$ sections s.t. the non-zero elements of $\zeta_i(x)$ form an orthonormal basis $\forall x \in X$, and s.t. $\zeta_i = 0$ is measurable $\forall i \in I$. A section s is **measurable** if

- $\forall i \in I$, $x \mapsto \langle s(x), \zeta_i(x) \rangle$ is measurable, and
- $x \mapsto \langle s(x), \zeta_j(x) \rangle$ vanishes outside a countable $I_s \subset I$.

Let μ be a measure on $\mathcal{A}$. A measurable section is a **nullsection** if it vanishes outside a set of measure zero. The **discrete Hilbert integral** is the vector space of measurable sections s for which $\|s\|^2 := \int_X \|s(x)\|^2 d\mu(x) < \infty$,

modulo nullsections. It is denoted by $H = \int_X H_x d\mu(x)$ and has an inner product

$$\langle s, t \rangle = \int_X \langle s(x), t(x) \rangle d\mu(x).$$

vanishing basis

We have a unitary isomorphism $H \rightarrow \hat{\bigoplus} L^2(X_i)$, where $X_i = X \setminus V(\zeta_i)$, given by

$$s \mapsto (x \mapsto \langle s(x), \zeta_i(x) \rangle).$$

Representations η_x of G on H_x s.t. $x \mapsto \langle \eta_x(g) \zeta_i, \zeta_j \rangle$ is measurable $\forall g \in G, i, j \in I$ define a representation $G \otimes H$.

Let A be locally compact abelian. Then G acts on the discrete Hilbert integral of the 1-dimensional Hilbert integral $H \rightarrow \hat{A}$ by $\eta_x = X$. Then $\int_X H_x d\mu \cong L^2(\hat{A})$.

Plancherel's Theorem

A group G is called a **type I group** if every factor representation is of type I, i.e. quasi-equivalent to an irr. rep.

Examples:

- Abelian groups ($VN(\pi)$ is commutative);
- Compact groups (unitary reps are direct sums of irreps);
- Nilpotent and semisimple Lie groups;

Lemma: Let G be separable. Then every irrep has separable Hilbert space.

We have $\hat{G} = \bigcup_{n=1}^{\infty} \hat{G}_n$ with the union σ -algebra of the smallest σ -algebras making $\pi \mapsto \langle \pi(g) v, w \rangle$ measurable.

hence separable

Thm: Let G be second countable, unimodular, locally compact, and of type I. Then $\hat{G}$ has a unique measure μ s.t. $\forall f \in L^1(G) \cap L^2(G)$,

$$\|f\|_2^2 = \int_{\hat{G}} \|\pi(f)\|_{HS}^2 d\mu(\pi).$$

by left and right multiplication

The map $f \mapsto (\pi \mapsto \pi(f))$ extends to a unitary $G \times G$ -equivariant map

$$L^2(G) \xrightarrow{\sim} \int_{\hat{G}} HS(V_\pi) d\mu(\pi).$$

Note: This is a generalisation of the Plancherel theorem in the abelian case, and the Peter-Weyl theorem in the compact case.

The Selberg Trace Formula

Prop: Let G be locally compact. If G has a unimodular closed cocompact subgroup then G is unimodular.

Let $G > \Gamma$ be discrete. If G/Γ has an invariant Radon measure of finite volume then Γ is a **lattice**. A cocompact lattice is called **uniform**.

Note that discrete cocompact subgroups are uniform lattices.

Prop: If G has a lattice Γ then G is unimodular.

Thm: Let $G > H$ be unimodular, closed, cocompact. Then

$$L^2(H \backslash G) \cong \bigoplus_{\pi \in \hat{G}} N_H(\pi) \pi,$$

where $N_H(\pi) \in \mathbb{N}$.

An L^2 -kernel $k: X \times X \rightarrow \mathbb{R}$ is **admissible** if $\exists g \in C(X) \cap L^2(X)$ s.t.

$$|k(x, y)| \leq g(x)g(y).$$

An operator $S: L^2(X) \rightarrow L^2(X)$ is **admissible** if it is represented by an admissible kernel.

$f_0 \in L^1(G)$ for some compact unit vol'd U

Theorem (Trace Formula): Let G be first countable, locally compact, $\Gamma \subset G$ a uniform lattice. Let $f \in C_{\text{unif}}(G)^2$. For every $\pi \in \hat{G}$, $\pi(f)$ is of trace class, and

$$\sum_{\pi \in \hat{G}} N_\pi(\pi) \text{tr} \pi(f) = \sum_{\gamma \in \Gamma} \text{vol}(\Gamma_\gamma \backslash G_\gamma) \psi_\gamma(f).$$

conjugacy classes in Γ

orbital integral

$$\int_{G_\gamma} f(x^{-1}\gamma x) dx$$